Srikrishnaa Vadivel

Electrical Engineer | Photonics | Embedded Systems | Scientific Computing

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Summary

Electrical engineer with Cornell ECE training, photonics communication, embedded systems delivery, FPGA/MCU experience, and current Yale PhD work in computational materials and high-performance scientific software.

Experience

Yale University, PhD Researcher / Embedded Systems TA 2021–present

- Research first-principles self-trapped excitons using HPC systems and scientific software stacks.
- Teach and support embedded systems concepts spanning firmware, instrumentation, and debugging.

Probe Technologies / Weatherford, Software and Embedded Engineer 2019–2021

- Developed embedded sensor logging firmware and Lattice FPGA logic for geological well tools.
- Built CUDA/OpenGL acoustic waveform processing and visualization tooling.
- Contributed ASP and Microsoft database software for field data workflows.

FindLight, Photonics Blogger 2018–2019

- Wrote clear technical content on photonics products, optics, and applications.

Projects

- Verilog MIPS CPU and matrix-multiply accelerator with Icarus test benches.
- Meep ring resonator FDTD model for integrated photonics simulation.
- DOLFINx PDE notes and solver scaffold for finite-element methods.
- Robotic hand embedded project published in Circuit Cellar.

Education

Yale University, PhD Electrical Engineering, 2021–present
Yale University, MPhil Electrical Engineering, 2023
Yale University, MS Electrical Engineering, 2023
Cornell University, MEng Electrical and Computer Engineering, 2017–2018
Cornell University, BA Electrical and Computer Engineering, 2013–2017

Skills

Circuits, embedded C, Verilog, FPGA, MCUs, photonics, signal processing, OpenGL, CUDA, Python, C++, Linux, HPC, numerical methods.